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MANGALORE INSTITUTE OF TECHNOLOGY & ENGINEERING

(A Unit of Rajalaxmi Education Trust®, Mangalore)
Autonomous Institute affiliated to VTU, Belagavi, Approved by AICTE, New Delhi
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Model Question Paper

Department: **Artificial Intelligence and Machine Learning**
Course Title: **Machine Learning Operations** Course Code: **23AIPE312**
Duration: **3 Hours** Semester: **05** Section: **AIML** Date: **—**
Name of the Faculty: **Manjunath Prasad H. R.** Max. Marks: **100**

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M: Marks, L: RBT (Revised Bloom's Taxonomy) level, C: Course Outcome.

Q. No.	Question	M	L	C
1a	Define MLOps. Explain the three key reasons why managing machine learning life cycles at scale is challenging.	08	L2	CO1
1b	Explain the challenges of iterating ML models on the edge with suitable examples.	06	L2	CO1
1c	Explain why MLOps is essential for scale.	06	L2	CO1
	OR			
2a	Explain model monitoring from the perspectives of the DevOps team, the data scientist and the business.	08	L2	CO1
2b	Differentiate between data governance and process governance in MLOps.	06	L2	CO1
2c	What key questions must be answered when finding data to build ML models, including data governance constraints?	06	L2	CO1
3a	What is a machine learning model? Explain it in theory and in practice with examples.	08	L2	CO2
3b	Explain the data exploration process carried out before training a model.	06	L2	CO2
3c	How does feature selection impact MLOps strategy? Explain automated feature selection and feature stores.	06	L2	CO2
	OR			
4a	Explain how ML models are evaluated and compared, including choice of metrics and cross-testing.	08	L2	CO2
4b	Differentiate between high-bias and high-variance models. Explain stratified modelling with an example.	06	L2	CO2
4c	Explain explainability of individual predictions and why deep learning models are called black-box models.	06	L2	CO2
5a	Explain machine learning security, covering adversarial attacks, poisoning and other vulnerabilities.	08	L2	CO3
5b	Explain reproducibility and auditability in MLOps.	06	L2	CO3
5c	What is an ML artifact? Explain its contents and the testing pipeline.	06	L2	CO3
	OR			
6a	Explain CI/CD pipelines for machine learning with an example pipeline.	08	L2	CO3
6b	Explain the challenges of scaling ML deployments. Differentiate scalable and elastic systems.	06	L2	CO3

6c	Explain the maintenance measures for a model in production. How are canary releases related to A/B testing?	06	L2	CO3
7a	Explain input drift with the wine quality example. Describe the common causes of data drift.	08	L2	CO4
7b	An e-commerce company collects personal data from visitors in the European Union to train its recommendation model. Apply the GDPR and CCPA data privacy regulations to this case: identify the rights these visitors hold over their data, and explain what the company's ML pipeline must support when a visitor asks to be forgotten or objects to automated decision-making.	06	L3	CO4
7c	Differentiate statistical and business metrics used in ground truth monitoring. Explain the role of p-values.	06	L2	CO4
	OR			
8a	Explain the feedback loop in MLOps and the information recorded by the logging system, with a neat diagram.	08	L2	CO4
8b	An online retailer has built a new advertisement model that predicts whether a user will click an ad, but the business evaluates it on the buy rate. Justify why A/B testing, and not champion/challenger, must be used to compare it with the current model, and plan the A/B test by stating what must be decided before, during and after the test.	06	L3	CO4
8c	Explain the emergence of Responsible AI and its components.	06	L2	CO4
9a	Explain the business use case and model development for consumer credit risk management.	08	L2	CO5
9b	A bank's personal-loan scoring model, used for loans of three years, was trained before an economic recession. Apply the monitoring methodology for consumer credit risk models to plan how the bank should check whether the model is still reliable, naming the data drift and performance drift metrics it should use.	06	L3	CO5
9c	Differentiate push and pull channels for recommendation engines.	06	L2	CO5
	OR			
10a	Explain the characteristics of power systems that make consumption forecasting important, with a neat diagram.	08	L2	CO5
10b	An online electronics store's recommendation engine predicts the probability that a customer clicks a recommendation. After launch, sales volume rose but overall revenue fell. Apply the textbook's guidance to explain the likely cause, choose a suitable KPI with its evaluation period, and design the control and experimental groups for testing a new strategy.	06	L3	CO5
10c	Differentiate the SCADA system and the metering system used in power grids.	06	L2	CO5